Accuracy Computation

4 Module Scenes — Panelist Brief (Simple Explanation)

1. Overview

Each module defines accuracy based on the skill it trains. Three modules use a simple success rate. Pen Trace measures path-following closeness.

Module	What "correct" means	Core formula
Ball Drop	Catching the ball	Caught ÷ Spawned × 100%
Color Sort	Correct color placement	Correct ÷ Total placements × 100%
Reaction Light	Clearing a light set	Sets cleared ÷ Sets played × 100%
Pen Trace	Ink closeness to guide path	Average pattern closeness scores

2. Ball Drop

Idea: Of all balls that appeared, how many did the player catch?

```
Accuracy = (balls caught ÷ balls spawned) × 100%
```

Example: 18 caught out of 20 spawned → **90%**

Also tracked (separate from main accuracy)

- **Dominant-hand accuracy** = dominant catches ÷ dominant balls
- **Non-dominant-hand accuracy** = non-dominant catches ÷ non-dominant balls
- **Hand difference** = dominant % − non-dominant %

Wrong grabs are counted separately and do not reduce the main accuracy percentage.

3. Color Sort

Idea: Of all pieces the player placed, how many went into the correct color?

```
Accuracy = (correct placements ÷ total placements) × 100%
```

Example: 24 correct out of 30 placements → **80%**

Every resolved piece counts as one placement. A correct placement increases the numerator. A wrong placement still increases the total, so accuracy goes down.

4. Reaction Light

Idea: Of all light sets the player faced, how many were cleared successfully?

```
Accuracy = (sets cleared ÷ sets played) × 100%
```

Example: 12 cleared out of 15 sets → **80%**

A set counts as cleared when finished correctly. Timed-out or failed sets still count as played, so they lower accuracy.

5. Pen Trace

Idea: This is not a simple right/wrong count. It measures how closely the player's ink follows the guide line.

Step A — Score one pattern

1. Sample the guide path every **1 cm**.
2. At each sample, measure distance from the guide point to the player's ink.
3. Convert that distance into a closeness score from 0 to 1:
   - Within **0.7 cm** → perfect (1.0)
   - From 0.7 cm to **1.5 cm** → score fades toward 0
   - Beyond 1.5 cm → 0
4. Average all sample scores, then multiply by 100 → **pattern accuracy %**

```
Pattern Accuracy = (average closeness of all samples) × 100%
```

Step B — Round accuracy (results screen)

```
Round Accuracy = (sum of finished pattern accuracies) ÷ (number of finished patterns)
```

Example: patterns scored 92%, 85%, 88%
Round accuracy = (92 + 85 + 88) ÷ 3 = **88.3%** (shown as ~88%)

The displayed accuracy average uses finished patterns only.

Rating feedback

- Below 60% → Fair
- 60% to 84% → Good
- 85% and above → Great

One-line takeaway for the panel:

Ball Drop, Color Sort, and Reaction Light use a percentage of successful attempts. Pen Trace uses a percentage of path-following closeness, then averages that across completed patterns.

Source: ModuleScenes accuracy logic in BallDropRound, ColorSortRound, ReactionLightRound, and PathManager (Pen Trace).